VibePrevisBench: Evaluating Agentic Coding for 4D Scene Directing in Blender

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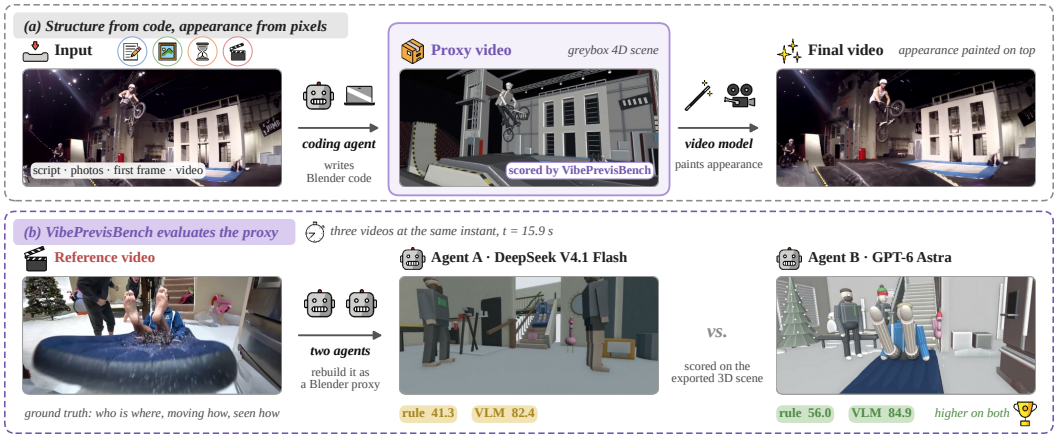


Figure 1. **VibePrevisBench scores the proxy, not the pixels.** (a) A coding agent turns a brief into a greybox proxy video; a video model paints the final video on top. (b) Two agents rebuild the same reference video, with their rule and VLM scores.

Abstract

Video generators can now follow a greybox proxy video that guides where people stand, what they do, and how the camera moves and cuts. Coding agents can build such proxies by writing Blender code, yet no benchmark evaluates how well they do it. We introduce VibePrevisBench, 242 tasks in which a coding agent turns a script, photos, a first frame or a real multi-shot video into an executable 4D scene. Each delivery is scored in two ways: program checks re-run the code and compare the exported scene with 84.4k hidden ground-truth facts on layout, actions, events, camera and cuts, and a VLM judge checks that these facts can be seen in the proxy video. We evaluate 11 frontier coding agents, and the best scores only 77.7/100. We find that agents differ most on tasks with image or video input, and that they all make the same mistakes: they get camera moves such as tilts and zooms wrong, leave out background objects the brief mentions, and make people walk too far. Nearly every run inspects its own renders, yet runs that view more renders do not score higher. We release the benchmark data and evaluation code, and anticipate that VibePrevisBench will open new pathways toward coding agents that reliably direct the scenes video generators bring to life.

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1 Introduction

Filmmakers have long rehearsed a shot before filming it: previsualization (previs) blocks out actors, camera and cuts with rough 3D models [41, 58]. Video generation now works the same way. Commercial models such as Kling and Seedance offer greybox (white-model) control [3, 32]: the user uploads a greybox render of the scene, a proxy video, that fixes where people stand, what they do and how the camera moves and cuts, and the model paints the final footage on top. Conditioning on such proxies makes layout, motion and camera far more controllable [4, 21, 31, 36, 43]. But building a proxy, like building previs, takes skilled 3D artists and time [24, 65].

Coding agents can now build it instead. Given a script, a few photos, a first frame or real footage, the agent writes Blender code, runs it, renders previews, inspects its own frames and revises [8, 9, 11, 33, 34, 36]. The result is an executable 4D scene, so moving an actor or a camera is a one-line edit and a re-render rather than another round of prompting and sampling. ***The agent becomes the director; the video model, the renderer.***

Whether the agent directs correctly matters, because a generator copies whatever its proxy gets wrong. We give Seedance 2.5 the same script with the proxy of a good and of a bad agent (Fig. 2). When the script asks the camera to crane up from eye level to a high angle, only the clip made with the good proxy cranes up; the bad proxy keeps its camera still, and so does its clip (Fig. 2a). On a second script, the bad proxy builds a person as a bare cylinder and places its camera inside a wall; the clip turns the cylinder into a man without legs and films a blank wall (Fig. 2b). Unlike pixels, a proxy can be checked directly, since its entities, cameras and cuts are explicit.

Existing benchmarks do not check the proxy. Video-generation benchmarks score pixels, not layout, event timing or camera paths [13, 28, 49, 68]. The concurrent PROWBench checks whether video models render what a program specifies [29]; it evaluates the renderer, whereas we evaluate the director. Video-understanding benchmarks ask sampled questions or score free-text captions, which a single frame or language priors can often answer [6, 17, 37, 52, 60]. Code-to-3D benchmarks mostly cover single objects, edits, static scenes or short dynamic clips [14, 19, 20, 46]. The closest works rebuild videos as 3D scenes but take only video as input and score rendered frames, not the scene [51, 53]. None asks an agent to direct a long, multi-person, multi-shot clip from a script, photos or footage and checks the result against ground-truth scene state (Tab. 1).

We introduce **VibePrevisBench**, 242 tasks that ask whether a coding agent can turn a brief (a script, photos, a first frame or a reference video) into an executable 4D scene that can serve as the proxy for a video generator. In each task, the agent writes and runs Blender code in a sandbox and delivers a greybox scene with its proxy video: who stands where, who moves when, how every shot is framed and where it cuts. Briefs come in four kinds (Fig. 3). **Scripts:** places, people, timed actions and shots are described in prose, which the agent must turn into metric 3D, timing and cameras. **Photos:** one to five photos of a real place with measured 3D ground truth; the agent rebuilds it at real scale and matches each photo's camera. **Continuations:** a first frame and a description of the next seconds, to be matched and then staged. **Reference videos:** real clips of up to 60 s to be rebuilt as 4D scenes, with layout, action timing and the camera of every shot read from the footage.

We score each delivery in two steps (Fig. 5). First, we re-run the delivered code in a fresh sandbox (a delivery that does not run scores zero) and export the scene state frame by frame. **Program checks** (the rule score) compare it with 84.4k hidden ground-truth facts on layout, actions, events, camera and cuts, such as whether a person reaches the door in the right second. Second, a **VLM judge** watches the proxy video and checks that what passed is visible, e.g., that the person is not hidden behind a wall. The overall score is their mean.

We evaluate 11 frontier models as coding agents (Sec. 5). The best, GPT-6 Astra, scores 77.7, and on only 35 of the 242 tasks does any agent reach a rule score of 90. **First**, agents differ

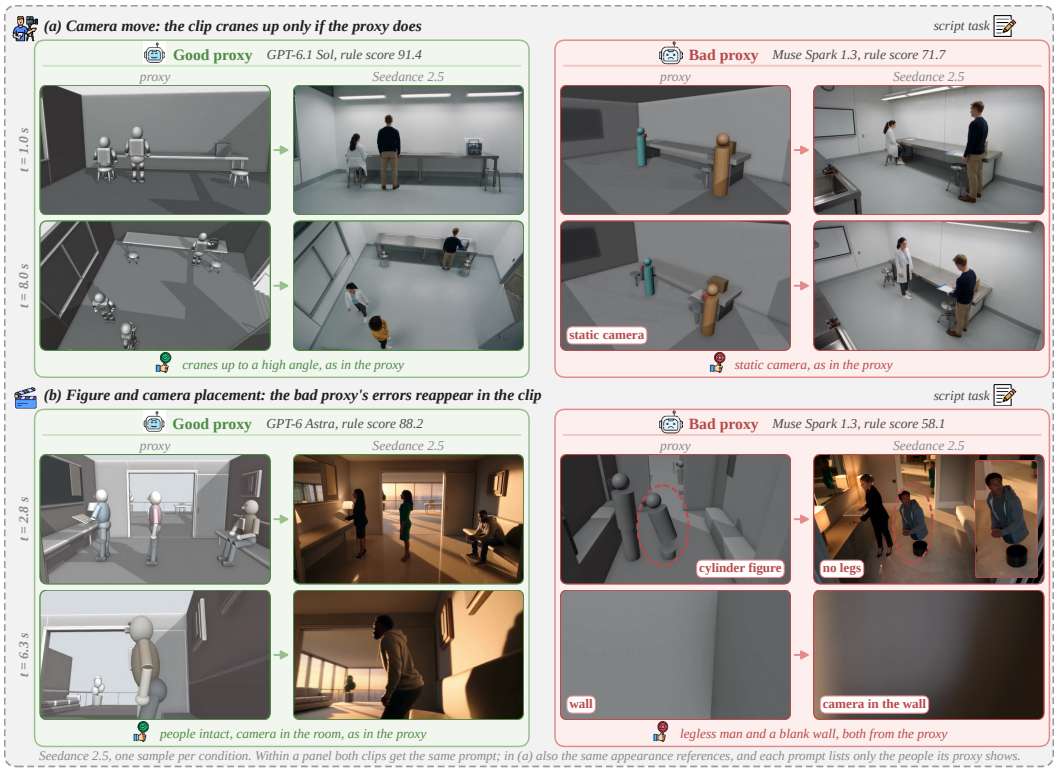


Figure 2. **Same prompt, different proxy: the generator copies whatever its proxy gets wrong.** For two script tasks, a video generator renders the same prompt with the proxy of a good and of a bad agent; in (a) both clips also share the appearance references, and each prompt lists only the people its proxy shows. (a) The script asks for a crane up to a high angle: only the good proxy cranes up, and so does only its clip. (b) The bad proxy builds a person as a bare cylinder and puts the camera inside a wall; its clip shows a man without legs and a blank wall.

most when they must read visual input: the best and worst agents are 12 points apart on scripts, but 17 on videos and 20 on photos. **Second**, the errors are systematic and shared by all eleven agents: 35–66% of low-angle shots are made eye level, 75% of tilts and 57% of zooms are wrong, up to 64% of video runs add cuts, set dressing mentioned only in the text is missing three times as often as listed objects (35% vs. 11%), and every agent's paths run 14–22% too long. **Third**, agents look but do not fix: nearly every run inspects its renders, yet for the same agent, runs that view more renders do not score higher.

In summary, our contributions are:

- **A benchmark for agentic previs.** VibePrevisBench has 242 tasks over script, photo, continuation and video inputs, with long, multi-person, multi-shot targets. Agents deliver an executable 4D scene that can serve directly as the proxy for a video generator.
- **A validated scene-state evaluation.** A rule scorer checks the exported scene fact by fact against ground truth, and a VLM judge checks that what is correct can be seen on screen. We validate them with injected defects, re-scoring and re-runs, three VLM judges and a human preference study; the agent ranking does not depend on the judge.
- **A large-scale study of frontier agents.** We benchmark 11 frontier agents, identify where agents fail (camera angles, cuts, text-only set dressing), and probe the role of perception with an oracle-description ablation and an analysis of how agents inspect their renders.

Table 1. **Comparison with related benchmarks.** *Duration*: typical target length in seconds (ours: median output length, as in Fig. 11; ∼: estimated; –: static or not fixed). *4D scene*: the model builds an animated 3D scene. *GT check*: the built scene is checked against ground-truth scene state. *Checks/task*: scored items per task (ours: median rule checks, from 84.4k ground-truth facts; Fig. 11). BVB reuses VSI-Bench's videos and questions. ✓ yes, ○ partial, ✗ no.

	Input			Scale		People and action			Filming		Output	Evaluation	
Benchmark	Text	Image	Video	#Tasks	Duration (s)	People	Interact.	Timed act.	Camera	Multi-shot	4D scene	GT check	Checks/task
Video generation													
VBench-2.0 [68]	✓	✗	✗	–	–	✓	○	✗	○	✗	✗	✗	–
WorldScore [13]	✓	✓	✗	3k	–	✗	✗	✗	✓	✗	✗	✗	–
FilmBench [55]	✓	✗	✗	1.2k	–	✓	○	✗	○	✓	✗	✗	–
Video understanding													
VSI-Bench [60]	✗	✗	✓	288	∼100	✗	✗	○	✗	✗	✗	✗	18
CameraBench [37]	✗	✗	✓	3k	5.7	○	✗	✗	✓	✗	✗	✗	–
VDC [6]	✗	✗	✓	1k	28	✓	○	✗	○	✗	✗	✗	–
Code to 3D / 4D scenes													
3DCodeBench [19]	✓	✓	✗	212	–	✗	✗	✗	✗	✗	○	✓	–
BlenderWorld [14]	✓	✓	✓	223	–	○	○	✗	○	✗	○	○	12
Code4Scene [63]	✓	✓	✗	190	–	✗	✗	✗	✗	✗	○	○	–
CinemaBench [66]	✓	✗	✓	810	∼10	○	✗	✗	✓	✗	✗	✗	–
4DCodeBench [46]	✗	✗	✓	200	∼7	✗	✗	○	✗	✗	✓	○	6
BVB [51]	✗	✗	✓	288	∼100	✗	✗	✗	○	✗	○	✗	18
CineRecon [53]	✗	✗	✓	50	15	✓	✓	✓	✓	○	✓	✗	193
CodeVideoBench [36]	✓	✗	✗	22	5	○	○	✗	✓	✗	✓	✗	–
VibePrevisBench (ours)	✓	✓	✓	242	40	✓	✓	✓	✓	✓	✓	✓	**408**

2 Related Work

Proxy-conditioned generation and previs. Video generators condition on cameras [1, 22, 57], 3D boxes and trajectories [18, 54, 70], coarse 3D and its renders [21, 45, 61, 69] and unified control signals [4, 31, 43]. Previs systems put rough 3D blocking in front of a generator [12, 24, 33, 42, 65], and agentic systems write that blocking as code [8, 9, 16, 34, 36]. Benchmarks on this side test the renderer, e.g., whether a video model renders what a program specifies [29, 44]. All use a proxy; none measures how well an agent writes one.

Evaluating generated video. Quality, composition and physics benchmarks [2, 13, 28, 39, 49, 68] score pixels with learned or VLM judges, which have their own blind spots [5]. Recent ground-truth checks cover short, single-shot physics or motion [7, 15, 30], and multi-shot benchmarks rely on judges or detected cuts [47, 55, 56]. None checks layout, event times, camera paths and cuts against ground truth; VibePrevisBench checks them in the proxy, where they are still explicit.

Agents that build 3D and 4D scenes with code. LLM agents write Blender code for objects, scenes, animation and films [23, 25, 27, 48, 59, 64]. Their benchmarks mostly target static content, from edits and objects to whole scenes [14, 19, 20, 63, 67]; give the agent a finished scene and ask only for cameras [35, 66]; or rebuild videos of physics, indoor scans or demonstrations [46, 50, 51]. The concurrent CineRecon [53] is closest: agents rebuild 50 cinematic clips as Blender scenes, scored by a VLM checklist on rendered frames. VibePrevisBench differs in three ways: most tasks start from a script, photos or a first frame; the clips are longer and multi-shot; and facts, including metric layout and routes, are read from the exported scene state (Tab. 1).

Video understanding and dense captioning. Video QA tests sampled facts [17, 40, 60], and camera motion remains hard for VLMs [10, 37]. Dense captions are becoming structured by shot and event [6, 52, 62] but are still scored as text, and writing a scene out explicitly exposes gaps that QA hides [26, 38]. A video task in VibePrevisBench asks for the most complete such description: one executable scene that must agree with every fact at once.

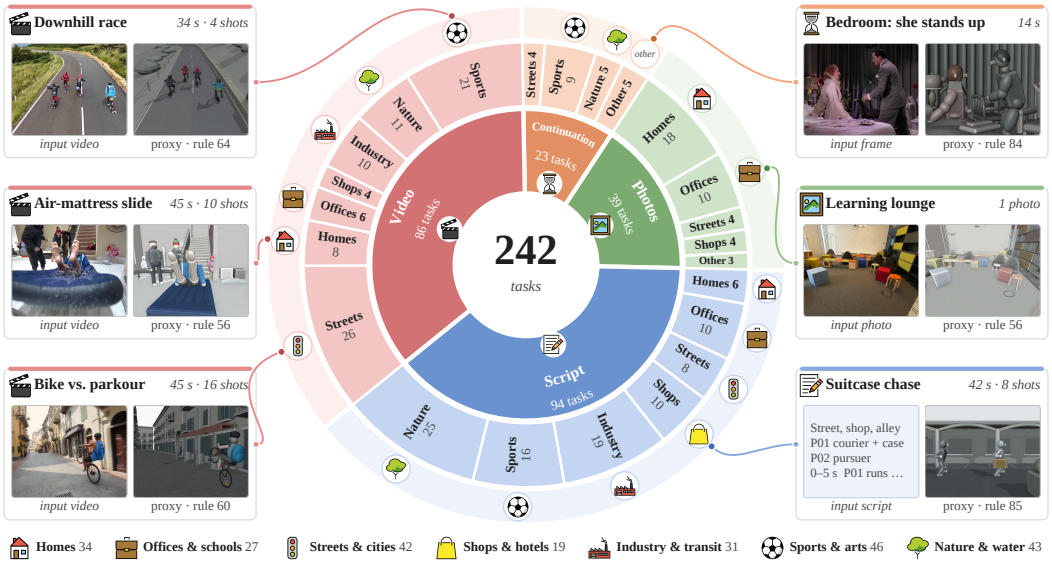


Figure 3. **VibePrevisBench at a glance: 242 tasks.** The inner ring gives the four input kinds, the outer ring the scene domains within each (spans proportional to task counts; domains with at most two tasks of a kind are merged into Other: photos Nature 2, Industry 1; continuations Homes 2, Offices, Shops, Industry 1 each); the legend row gives each domain's total. The 7 domains are those of Fig. 4. The six cards are real tasks, each with its input and GPT-6 Astra's proxy and rule score. One full example of each input kind is in Fig. 10; length and checklist distributions are in Fig. 11.

3 VibePrevisBench

Each task is a director's brief that an agent must turn into an executable 4D scene, and behind each brief sits a hidden ground truth broken into facts that can be checked one by one. We define the task (Sec. 3.1), describe the four kinds of input (Sec. 3.2), and explain how the tasks (Sec. 3.3) and their ground truth (Sec. 3.4) were built.

3.1 Task

A task gives an English brief, optional media (photos, a first frame or a reference video), a list of places and subjects with ids and short descriptions, and an output specification (frame count, frame rate, resolution). The agent writes a Blender program; executing it yields a time-varying scene of tagged entities and places, per-frame transforms, and cameras with their cuts, and rendering that scene yields the proxy video. The proxy is a greybox of boxes, simple solids and jointed mannequins in flat colors. Appearance is out of scope, since the video model supplies it. What is in scope is everything a director must fix before a frame is generated: which subjects exist, where they stand and which way they face; how the places connect; who moves where and when, and what they do; where each camera stands, how it frames and moves; and when the cuts fall. Subjects are referred to by their ids, so every fact points to one object.

3.2 Four kinds of input

Four kinds of input (Fig. 3; one example of each in Fig. 10) move from being told to having to see. **Scripts** are client-brief screenplays with several connected places, a timed beat sheet and a shot list. Layout is given in a speaker's frame ("stand in the garage facing the tunnel exit: the desk is against the wall on your left"), so the agent must turn prose into metric 3D, choreograph every subject on time, and block cameras that show each beat. **Photos** show a

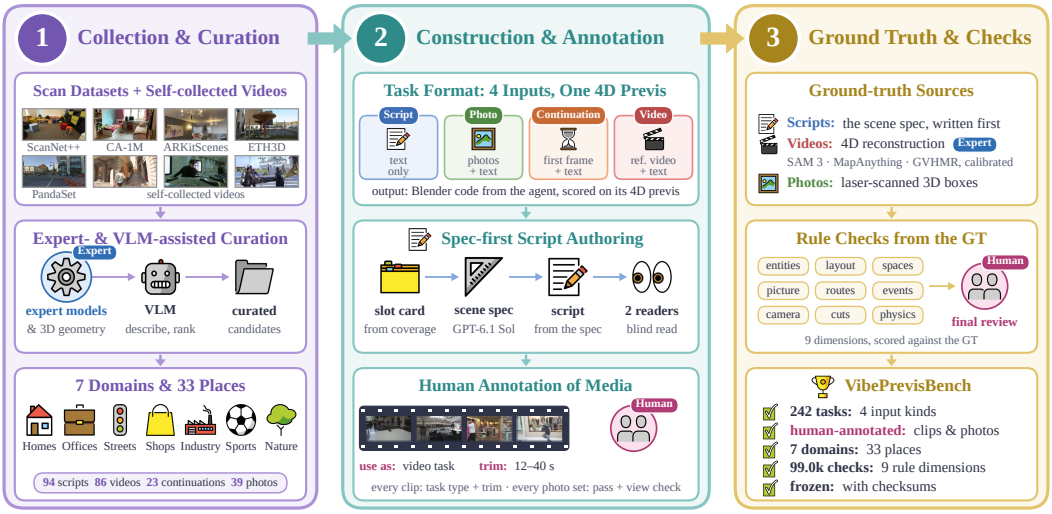


Figure 4. **How VibePrevisBench is built.** (1) Scan datasets and self-collected videos are curated with expert models and a VLM into 7 domains and 33 places. (2) Scripts are written spec-first and must survive two blind readers; clips and photos are annotated by humans. (3) Ground truth comes from the scene spec, 4D reconstruction or laser-scanned boxes, and is turned into checks in nine dimensions.

real place from one or a few views; the agent rebuilds it at real-world scale and places one camera per photo so that its view matches. This is single- or multi-view 3D reconstruction, expressed as a program. **Continuations** give a first frame and a description of the next seconds; the agent must match the frame and then stage the action. **Reference videos** are real multi-person, multi-shot footage. The brief lists only the places and subjects with their ids; layout, who does what and when, and the camera of every shot must all be read from the video, and the output must start at its first frame and cut where it cuts.

3.3 Construction

We build tasks for coverage and then prune them (Fig. 4); details are in Appendix A.1. Two taxonomies, of what a scene holds and of how a task is asked, set coverage targets, and gaps in the pilot tasks become new tasks in three lanes. For scripts, the ground truth is written before the script, and at least one of two blind readers must recover every scored claim. Clips for video and continuation tasks are retrieved, pre-screened and chosen by human reviewers. Photo tasks use only scans with measured 3D boxes and camera poses, and the boxes are checked against every photo. Finally, we drop tasks that do not discriminate between agents (by item-rest correlation) or have source problems, leaving 242.

3.4 Ground truth

Scripts. The ground truth is written first and every claim is anchored to a sentence of the script and survives the blind read (Sec. 3.3). **Photos.** The ground truth is the source datasets' measured 3D boxes and camera poses, with no model in the loop; moving or unconfirmed vehicles in street scenes are excluded. Photo tasks are the only kind whose ground truth is measured. **Continuations and videos.** The ground truth is estimated from the footage. We detect the shots; register every subject over the whole clip with Gemini 3.8 and observe each shot in detail with Gemini 4; segment subjects with SAM 3; estimate depth and per-frame cameras with MapAnything and VGGT, keeping shots that pass consistency checks; recover bodies with GVHMR; lift and register everything in 3D; and have Gemini 4 re-check disputed items and extract events with tolerance windows (Fig. 9). The output is a ground-truth scene and a checklist of time-anchored facts; facts the pipeline cannot ver-

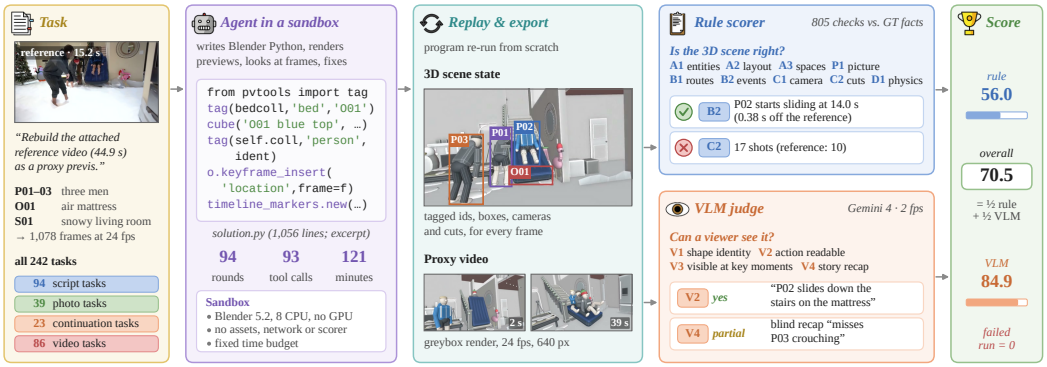


Figure 5. **One task through the evaluation pipeline** (video task M50_049, solved by GPT-6 Astra). The agent writes Blender Python in a sandbox, rendering previews and inspecting its own frames. We re-run the delivered program from scratch, export the tagged 3D scene state for every frame and render the greybox proxy. The rule scorer checks the scene against ground-truth facts (two of 805 checks shown); the VLM judge watches the proxy and is asked only about items the rules did not fail (two verdicts shown). Overall is the mean of the rule and VLM scores; a failed run scores 0 on both.

ify are marked unscorable and never count. Tasks, media, source manifests, ground truth and the scorer are versioned and frozen together with checksums; the composition of the benchmark is summarised in Fig. 3, and Fig. 11 gives per-kind statistics: prompt and output lengths, rule checks per task, and the share of tasks with each scene demand.

4 Evaluation

The program score asks whether the scene is right; the VLM score asks whether what is right can be seen. Fig. 5 follows one task through the four steps: the agent directs the scene, the delivery is validated, we replay and export it, and the two scores are computed. The overall score is $\frac{1}{2}$ rule + $\frac{1}{2}$ VLM, both on 0–100, averaged over tasks with equal weight. A run that delivers no valid scene scores 0 on both rather than being skipped, so no agent gains by failing hard tasks.

4.1 Directing, delivery and replay

Each run takes place in a sandbox with Blender 5.2, eight CPU cores and no GPU, no external assets or network, no access to the scorer, and a fixed time budget of 4 hours. The agent writes and runs Blender code, renders previews, inspects its own frames and revises. It tags entities and places with the task's ids through a small toolkit, so every fact binds to an object without visual matching. A delivery must pass three gates of 21 automatic checks: the code runs standalone, the video is rendered in full at the specified resolution, frame rate, duration and codec, and the scene reopens in Blender with the task's settings. Otherwise the run scores 0. We then re-execute the delivered program from scratch in a fresh sandbox, dump the scene state frame by frame (tagged entities and places, their oriented boxes and parts, cameras and cuts) and render the greybox proxy at 24 fps. Nothing the agent rendered itself is scored, so a good-looking video cannot stand in for a correct scene.

4.2 Rule score on the exported scene

The scorer first aligns the agent's coordinate frame to the ground truth with a similarity fit on anchored bodies. It then checks the hidden ground-truth facts; the 84.4k scorable facts expand into 99.0k rule checks (408 per task at the median), since one fact can be checked in several ways, e.g. the order of every pair of events. The checks are grouped

Table 2. **Main results on 242 tasks.** The rule and VLM scores are given per input kind and over all tasks (*All*); the VLM score uses the Gemini 4 judge and, on photos, judges only whether subjects and places can be recognised (V1); *Overall* is $\frac{1}{2}$ rule + $\frac{1}{2}$ VLM. Failed runs score 0. Per column, the best score is in **bold** on a darker shade and the second best on a lighter shade. Reasoning effort in grey.

Agent	Rule					VLM					Overall
	Script	Photo	Cont.	Video	All	Script	Photo	Cont.	Video	All	
GPT-6 Astra max	**85.6**	55.8	**79.8**	**56.8**	**70.0**	**76.5**	93.6	**91.1**	**89.8**	**85.4**	**77.7**
GPT-6.1 Sol max	84.4	55.3	78.1	56.0	69.0	73.3	**93.8**	89.5	87.3	83.1	76.1
Gemini 4 Argon high	82.1	57.1	78.5	55.8	68.4	72.7	90.3	88.8	85.6	81.7	75.0
Opus 5.5 high	83.0	**59.8**	78.2	53.9	68.4	73.0	90.4	82.5	81.6	79.8	74.1
GPT-5.6 Sol max	80.8	44.2	73.0	49.0	62.9	69.4	92.5	81.1	78.7	77.5	70.2
DeepSeek V4.1 Flash max	80.1	43.2	73.0	45.7	61.3	70.8	90.2	82.4	78.8	77.9	69.6
Kimi K3 xhigh	81.3	42.4	73.9	45.2	61.5	69.3	81.2	80.9	72.5	73.5	67.5
Gemini 3.8 Flash high	78.6	43.1	71.4	46.2	60.7	59.1	87.7	77.6	68.3	68.7	64.7
GLM 5.3 Flash high	76.0	39.9	71.0	42.6	57.8	61.0	84.8	76.3	70.2	69.5	63.7
Doubao Seed 2.1 Pro max	73.6	40.4	65.7	42.2	56.3	56.1	86.8	69.1	67.2	66.2	61.3
Muse Spark 1.3 xhigh	77.4	40.5	67.2	40.1	57.2	58.3	84.7	72.5	61.3	65.0	61.1

into nine dimensions (Tab. 3 in the appendix), each within a registered tolerance and with partial credit; for example, a run whose P01 starts running 0.46 m from the reference start point scores 0.97 on that fact. A block of facts with pass rate P and chance rate R scores $\text{clip}((P-R)/(1-R))$ times the coverage of the expected entities, so guessing earns nothing and an empty scene cannot pass relational facts vacuously. Physical validity (D1) is multiplied by $\min(1, E/0.25)$, where E is the mean score of layout, routes and events, so a near-empty scene cannot collect physics points. The rule score is the mean of the dimensions a task exercises, times 100. The scorer is deterministic.

4.3 VLM judge on the rendered proxy

A generator sees only pixels, so a correct structure that cannot be seen is of little use. The VLM judge therefore does not re-check what the rules measure; it asks whether what the rules accept can be seen on screen. Gemini 4 watches the proxy at 2 fps, together with the reference video for video tasks; for photo tasks it also watches an orbit render of the delivered scene, so that a matched photo view cannot hide an empty room. The judge never sees the model's name or the object names in its scene. In a first turn it receives no brief and retells the clip blind. In a second turn it receives the brief and checklists generated from the ground truth (key entities and places, events, key moments and story beats) and gives a verdict on each item: yes, partial or no, with a time span and a reason. Items the rules have already failed are not asked again. Four dimensions are judged: V1, whether subjects, objects and places can be recognised (yes only if the blind recap already named them); V2, whether each action reads on screen, given that its subject is visible; V3, whether the subject is visible and unobstructed at each key moment; and V4, whether the blind recap retells the story beats in order. Photo tasks are judged on V1 only. With yes = 1, partial = $\frac{1}{2}$ and no = 0, the VLM score is the mean share over the applicable dimensions, times 100. We do not ask for aesthetics, composition quality, pacing or acting.

5 Experiments

5.1 Setup

We evaluate 11 frontier models as coding agents. The GPT models run in the Codex CLI and all other models in Claude Code; each model's reasoning effort is given in grey in Tab. 2. Every task is run once, in a fresh sandbox with a fixed budget, and a run that delivers no

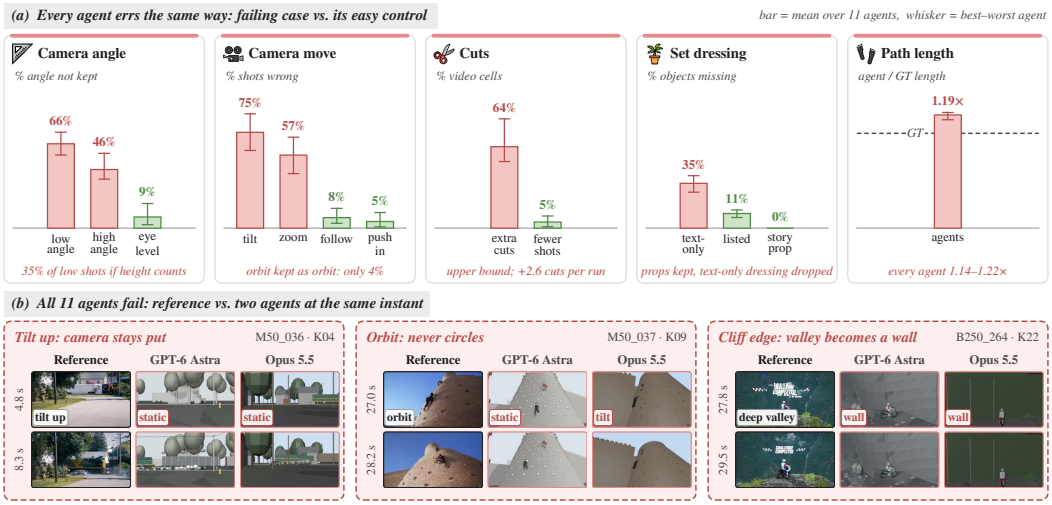


Figure 6. **Shared failure modes.** (a) Error rates on a hard case (red) and an easy control (green), averaged over the 11 agents; whiskers show the best and worst agent. *Camera angle*: low and high angles are made eye level. *Camera move*: tilts and zooms are mostly wrong, follows and push-ins rarely. *Cuts*: video runs add cuts. *Set dressing*: set dressing described only in the text is dropped, story props are kept. *Path length*: agent paths are longer than the GT. (b) Shots that the rules mark wrong for all 11 agents: the reference frame and two agents at the same instant; tags describe what the reference shows and what each agent produced.

valid scene scores 0; re-runs move the benchmark mean by only about 0.4 points (Tab. 4 in the appendix). Significance is tested with 10,000 paired bootstrap resamples of the tasks.

5.2 Main results

Table 2 ranks the agents. GPT-6 Astra is significantly first (77.7); GPT-6.1 Sol, Gemini 4 Argon and Opus 5.5 follow, and five of the ten adjacent pairs cannot be separated. The ranking is identical whichever of the three VLM judges enters the overall score (Appendix B.3), and a random 60-task subset already recovers it at Kendall $\tau \approx 0.95$. The benchmark is far from saturated: the leader is the best agent on only 91 of the 242 tasks, at least one of the bottom five beats it on 41 (Kimi K3 alone on 26), and on only 35 tasks does any agent reach a rule score of 90. Delivery is itself a capability: 14 runs produced no valid scene, including two by Gemini 4 Argon. Cost per task varies 45-fold, from $0.73 for GLM 5.3 Flash to $32.59 for GPT-6 Astra (Fig. 7). GPT-6.1 Sol reaches 98% of the leader's overall score at 21% of its cost, while Kimi K3, second in cost, ranks seventh.

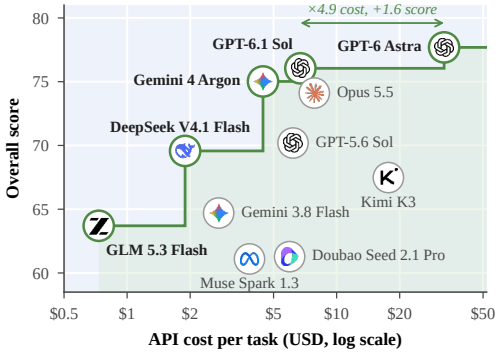


Figure 7. **Cost vs. score.** Mean API cost per task at list prices (log scale) against the overall score. The staircase is the cost–score frontier: no cheaper agent scores higher.

Input kinds exercise different dimensions (photos have no routes, events or cuts) and obtain their ground truth differently, so we compare agents within a kind. On videos, scores fall with clip length, shot count and moving shots (Spearman −0.27, −0.34, −0.38), but hardly with the number of people or objects. Replacing the reference video with a text description of its events, cuts and camera, generated from the ground truth, raises the score on all five tasks we test (+15.6 ± 6.0) at a quarter of the cost (Tab. 4 in the appendix), with gains in

routes, events, camera and cuts and none in layout. Much of the difficulty on these tasks thus lies in reading the reference.

5.3 Finding 1: Agents spread apart on visual input

Agents differ less when the brief is written out than when they must read it from images (Tab. 2). The best and worst agents are 12.0 points apart on scripts, 14.1 on continuations, 16.7 on videos and 19.9 on photos. On photos, whose ground truth is measured, agents differ most on the picture structure (P1, a spread of 46.8), about half as much on spaces, physics and the camera (A3 23.6, D1 22.9, C1 21.7), and least on the entities and the layout (A1 7.6, A2 8.2); on the layout every agent stays below 26. Better agents match the view better, but none of them rebuilds the room behind it.

5.4 Finding 2: The errors are systematic

Across inputs the errors point the same way for all eleven agents (Fig. 6). Agents build from the inventory: story props are never missing, but set dressing described only in prose is missing 35% of the time. Paths run 14–22% too long. Camera moves collapse: orbits are reproduced in only 4% of cases. Low-angle shots flatten to eye level 35–66% of the time, depending on whether camera height counts, and at most 64% of video runs add cuts the reference does not have. Some shots defeat every agent (Fig. 6b).

5.5 Finding 3: Agents look, but do not fix

Nearly every run inspects its own renders (13–40 images per run, Fig. 8a); at most 2% never look at one. Yet within an agent, once task difficulty is controlled, no process measure (turns, full renders, images inspected) predicts a better proxy: all median correlations lie within ± 0.06 (Fig. 8b). Across agents, more turns and more failed renders go with lower scores (Spearman -0.71 to -0.83, Fig. 15): weaker agents take longer. Finding 2's errors persist although every agent looks at its frames, so the bottleneck is noticing what is wrong and correcting it.

Further analyses (legibility and event completion, and the downstream effect on video tasks) are in Appendices C.2 and C.3.

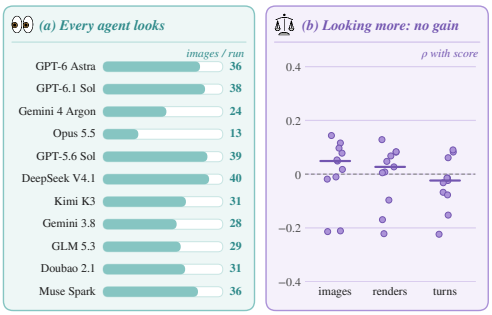


Figure 8. **Agents look, but looking more does not help.** (a) Images of its own renders an agent views per run, counted after its first render. (b) Within each agent, correlation of the score with all images viewed, full renders and turns, task difficulty controlled (dot = agent, bar = median).

6 Conclusion

We introduced VibePrevisBench, a benchmark for the director step of controllable video generation: a coding agent turns a script, photos, a first frame or real footage into an executable 4D scene, which we check fact by fact against ground truth and for what can be seen on screen. Across 11 frontier models, agents spread apart on visual input, make the same systematic errors, and look at their renders without correcting them. Because a video generator reproduces what its proxy gets wrong, closing these gaps is a direct path to video that can be directed, not just prompted.

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Appendix

A Benchmark details

A.1 Task construction

Coverage first. We defined two taxonomies, one for what a scene holds (places, spaces, layout, subjects, actions, camera and editing, genre) and one for how a task is asked (input form, how much information is given, length, writing style, organisation, noise, whether ids are given). We tagged the pilot tasks, required a minimum number of tasks per tag and per kind of place, and turned the gaps into slots for new tasks, routed to three lanes.

Scripts. Each slot becomes a card fixing genre, style, length, places, people, actions and a story seed. The ground truth is written before the script: a model (GPT-6.1 Sol) first codes the scene, then writes the script, then the claims, each citing the sentence it comes from. Automatic gates check the scoring of every task and audit the output specification. Two readers (DeepSeek models) who see only the script then try to recover every claim; a claim both misread is reworded and re-read for at most two rounds, and dropped if it is still misread.

Videos and continuations. We collected candidate clips ourselves, pre-screened them by cut, motion and people statistics and by model ratings of contact sheets (Claude, Gemini 4), and tagged the survivors by place, genre, people, shots and moving vehicles. Human reviewers decided whether a clip becomes a video task, a continuation or neither. Quotas then capped over-represented content such as vehicles, and borders and logos were cropped. Figure 9 walks through the ground-truth pipeline on one clip.

Photos. We drew only on scan datasets with measured 3D boxes and camera poses, indoors and in street scenes, and used each scene once. Ground-truth boxes were drawn back onto every photo and checked first by a model and then by a human; candidates whose boxes did not match the photo were rejected.

Pruning. All eleven agents ran the pool, and we dropped tasks that did not discriminate between agents: low item-rest correlation or little spread across agents, never a single agent's score. Finally we removed tasks with source or licensing problems or an unscorable run.

A.2 Task examples

Figure 10 shows one task of each kind; the prompts of the same four tasks are excerpted below.

Script (excerpt). *Suitcase Chase Through the Corner Shop. A courier is chased through a shop into a back alley, where a reversing van traps his accomplice… S02 Shop: north of the street's east part. A west shelf and an east shelf run north–south either side of the aisle from the front door to the back door… P01 person: courier holding O01, west of the planter, facing east. P02 person: pursuer, behind P01, facing east… 0–5 s: P01 runs east with O01, P02 chasing. 5–10.8 s: P01 enters the shop by the front door at 9.5 s; P02 follows at 10.8 s… Shots (lens: wide-angle 24 mm, moderately wide 28 mm, normal 35 mm, slightly long 50 mm)… 5–8 s, hard cut: under the arcade's east end; eye level, normal, facing west; pans right to the door…*

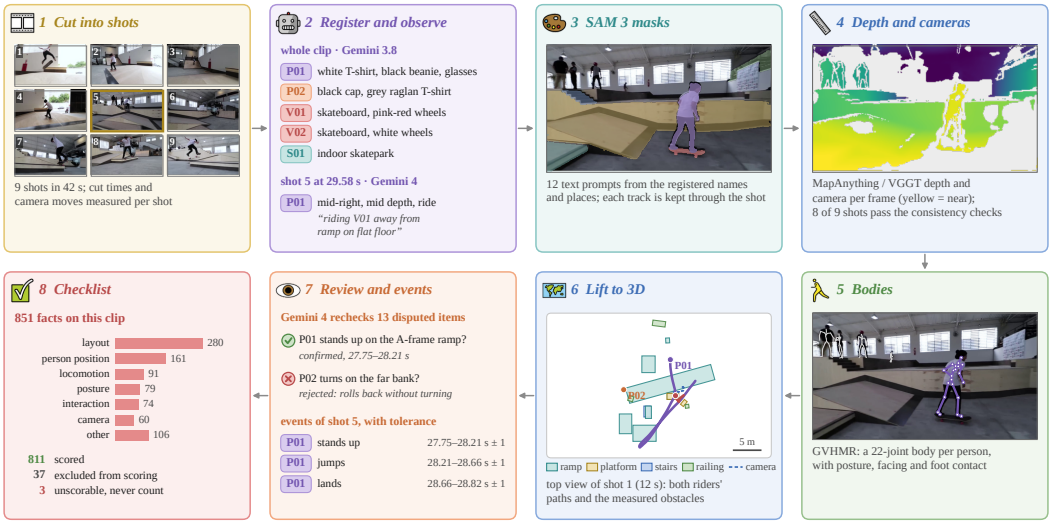


Figure 9. **Ground truth for a video task.** The eight steps of the video pipeline on one clip: shots, registration and per-shot observation, masks, depth and cameras, bodies, lifting to 3D, review and events, and the final checklist.

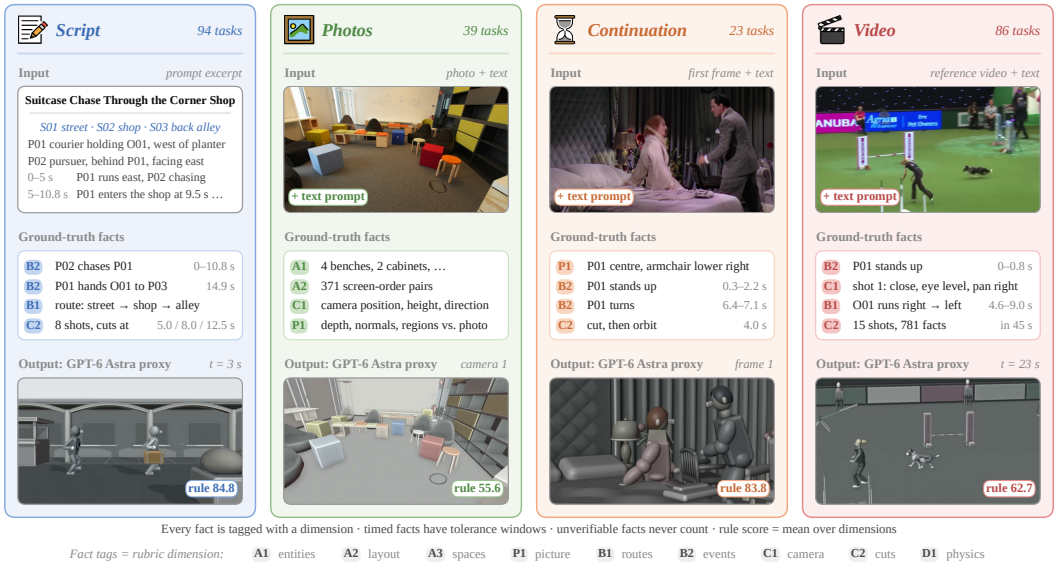


Figure 10. **The four kinds of input.** One task of each kind. *Top:* what the agent receives (a script excerpt, a photo, a first frame, a reference video); photo, continuation and video tasks also come with a text brief. *Middle:* some of the hidden ground-truth facts for that task, each tagged with its rubric dimension. *Bottom:* a frame of the proxy delivered by GPT-6 Astra, with its rule score. Timed facts carry tolerance windows, and facts the ground truth cannot verify never count.

Photos (prompt). *Rebuild the place shown in the attached photo as one static 3D proxy blockout, with one camera. Build the place and everything in it at real-world size, and place the camera so that its view matches the photo: position, direction and lens. Output: 12 frames at 24 fps, 640×426… Place: S01 Informal learning lounge. Tag every piece of the place (walls, floor, furniture, fixtures) by class. People in the photo are not part of the place: leave them out.*

Continuation (excerpt). *Continue the attached photo as a proxy previs. Frame 1 must reproduce the photo: the camera's position, direction and lens, and every visible subject and object where it is in the photo… Output: 336 frames at 24 fps (14.00 s)… P01 person: the seated woman beside the bed… P02 person: the standing man to her right… 0.0 s: the close eye-level side view of P01 and P02 pulls back in a dolly move. From 0.3 to 2.2 seconds, P01 stands up from the bed… 4.0 s: Cut to a medium eye-level front view of P01 with P02. The camera orbits toward her side… From 6.4 to 7.1 seconds, P01 turns back toward P02 and points firmly toward the armchair…*

Reference video (prompt). *Rebuild the attached reference video as a proxy previs. Match what the video shows: the places and what stands where, which way things face, how every listed subject moves and what it does when, and the camera of every shot (where it stands, how it frames and moves), with the cuts and transitions at the same times as in the video. Start at the video's first frame. Reference video: 45.02 s. Output: 1080 frames at 24 fps (45.00 s), 640×360. Places: S01 indoor agility arena. Subjects: P01 person: blonde woman with her hair tied in a ponytail… O01 dog: medium-sized dog with grey, black, and white fur…*

A.3 Task statistics

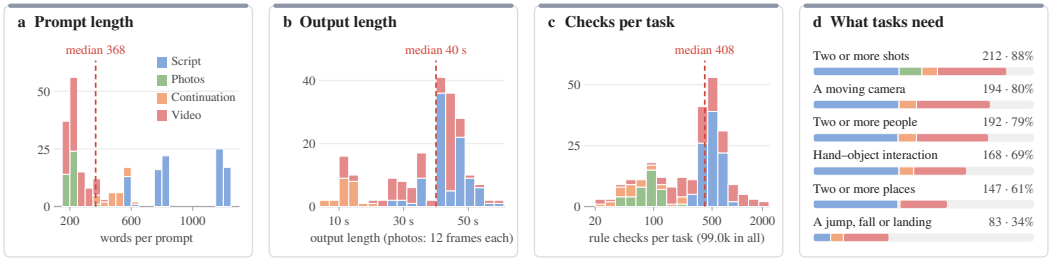


Figure 11. **Task statistics**, stacked by input kind: (a) prompt length; (b) output length (photo tasks render 12 frames per photo); (c) rule checks scored per task (log scale); dashed lines mark medians. (d) Share of tasks with each demand, read from the ground truth and split by input kind.

B Evaluation details

B.1 Rule-score dimensions

B.2 VLM judge rubric

The judge (Gemini 4) watches the proxy at 2 fps. Its prompt says that the video comes from a simplified previs, with simple shapes and mannequin-like figures in at most flat colours, so this look is not penalised. It never sees the agent's name or the object names in the scene.

Turn 1: blind recap. Without the brief, the judge describes only what it sees: in time order, who is where and what each person does, with people referred to by their look and objects by their shape ("unidentified shape" or "unclear movement" otherwise); what kind of place it is; and every object it can identify from its shape. For photo tasks it also watches

Table 3. **Rule-score dimensions.** The rule score is the plain mean of the dimensions a task exercises; P1 applies only to tasks with reference frames (photos, continuations and videos).

Dim.	What is checked against the ground truth
A1 Entities	every listed subject and prop exists, with the right class and count
A2 Layout	relative positions: sides, distances, depth and screen order, against walls
A3 Spaces	places, their extents and how they connect
P1 Picture	per-shot depth, normals, segmentation, edges and facing vs. the reference frames
B1 Routes	where each subject is over time, the order of places visited, direction
B2 Events	whether each event happens, when, in which order, with which posture and gait
C1 Camera	per shot: position, height, facing, lens, subjects in frame, camera move
C2 Cuts	number of shots, cut times, transitions, shot order and duration
D1 Physics	no interpenetration, floating or sinking, teleporting, implausible sizes

an evaluation-side orbit of the delivered scene (a 360° look from the room centre, then a top-down orbit with the ceiling hidden) and lists places and objects only.

Turn 2: verdicts. The judge then receives the brief (and, for video tasks, the reference video) and checklists generated from the ground truth; items the rules already failed are not asked. It is told that the automatic checks verify presence, positions, layout, camera placement, timing, shot count and physics, and that materials, lighting, realism, aesthetics and composition are not judged. Every entry gets *yes*, *partial*, *no* or *n/a* (the entity or shot is absent, which the rules already penalise) with the timestamps used, and the lower verdict when in doubt.

V1 Shape and place identity (key entities, places). *yes* only if the blind recap already named the category or kind of place; *partial* if it was not named blind but a shape at a plausible spot fits once the brief is known; *no* otherwise.

V2 Action readability (events). Would a viewer who has not read the brief describe the footage as that action? *yes*: action and contact are clear; *partial*: the pose changes but the action or contact is unclear; *no*: miming, standing still or another action. Events whose moment is not visible in V3 are left out of V2.

V3 Visibility at key moments. At each judged event, and in each shot without events, are the people and objects involved clearly visible? *partial*: small, partly blocked or partly out of frame; *no*: blocked, out of frame, too small, a blank or wall-only frame, the camera inside geometry, flicker or leftover helper objects.

V4 Story beats. Does the judge's own blind recap contain each story beat (*yes*: its gist; *partial*: part of it)?

With *yes* $= 1$, *partial* $= \frac{1}{2}$ and *no* $= 0$, each dimension is the share over its entries (*n/a* left out), and the VLM score is the mean over the applicable dimensions times 100; photo tasks use V1 only.

B.3 Is the evaluation trustworthy?

Determinism and direction. The rule scorer is deterministic: each of the 2,786 deliveries that were scored more than once during development received an identical score every time. On 12 high-scoring video deliveries, injected defects lower the rule score monotonically and in the intended dimensions: delaying all action by 4 s costs 9.8 points and keeping one person costs 12.1. A 90° camera yaw costs 3.6 points when only the exported cameras are turned, and 21.0 when the ray-cast picture buffers are also re-projected to the turned camera. The physics check misses floating people: lifting them by 1 m costs only 0.8.

Judges. Re-judged with GPT-6.1 Sol and Qwen3.8 Omni Flash, the agents' VLM scores correlate with Gemini 4's at Spearman 0.94–0.96, though only at 0.73–0.79 per run; Gemini 4 favours its own family by 1.3 points without changing any rank, and the overall ranking is identical whichever judge is used (Kendall $\tau = 1.00$).

Humans. On 242 pairs of proxies, one per task, an annotator picked the one easier to understand; each agent's human win rate correlates with its mean score from each judge at 0.90–0.92.

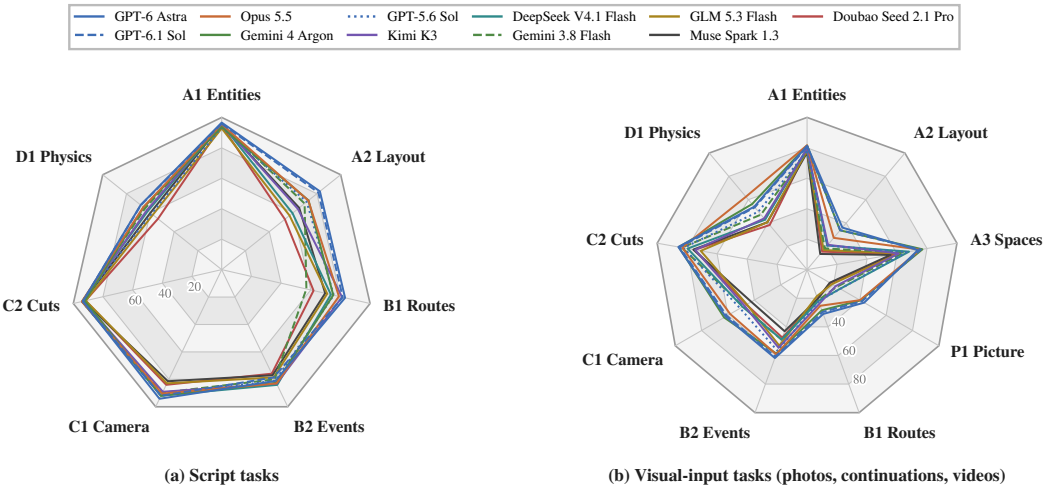
Noise. Re-running an agent changes a task's overall score by 5.5 points on average but the benchmark mean by about 0.4; five of the ten adjacent pairs are statistically tied.

C Additional results

C.1 Ablations and capability profiles

Table 4. **Ablations** on DeepSeek V4.1 Flash. Each row reruns the agent with one change and reports the paired per-task difference to the main run (mean $\pm$ s.d.). *Rerun*: the same tasks run again unchanged. *Oracle caption*: video tasks with the video replaced by a text description. $|\Delta|$: mean absolute change of the overall score per task. Cost: agent $ per task, reference $\rightarrow$ ablation. †The caption is generated from the ground truth (events, cuts, camera), so this row is an upper bound on what a perfect perception step would add.

Change	Tasks	n	ΔRule	ΔVLM	ΔOverall	$\lvert\Delta\rvert$	Cost
Rerun	Script	8	$-2.0_{\pm 5}$	$+2.5_{\pm 14}$	$+0.2_{\pm 7}$	5.5	0.57→0.70
Oracle caption†	Video	5	$+14.0_{\pm 8}$	$+17.2_{\pm 13}$	$+15.6_{\pm 6}$	15.6	1.94→0.47



(a) Script tasks

(b) Visual-input tasks (photos, continuations, videos)

Figure 12. **Capability profiles of all 11 agents.** Rule score per dimension (0–100, failed cells count 0) on (a) script tasks and (b) visual-input tasks. Exact values per agent are in Fig. 13.

C.2 Event completion and legibility

Correct is not always legible, and events separate agents most. The VLM judge is asked only about facts the rules accept. Of those, about a third of the actions cannot be read from the render (69% legible on average, from 86% for GPT-6 Astra to 51% for Muse Spark 1.3),

	A1 Entities [95.0]	A2 Layout [68.1]	B1 Routes [73.5]	B2 Events [80.5]	C1 Camera [89.0]	C2 Cuts [93.2]	D1 Physics [62.5]
GPT-6 Astra	**96.6**	**82.5**	**83.2**	81.2	**94.2**	93.6	**68.1**
GPT-6.1 Sol	95.8	81.2	81.6	78.6	92.0	93.6	68.0
Opus 5.5	95.5	72.8	79.0	83.0	90.4	**94.2**	65.9
Gemini 4 Argon	93.8	73.0	75.2	82.5	92.1	92.2	65.6
GPT-5.6 Sol	93.5	71.4	75.5	81.0	91.2	94.2	59.0
Kimi K3	95.9	64.0	81.1	83.2	89.1	93.1	62.8
DeepSeek V4.1 Flash	95.6	59.5	73.6	**84.0**	91.9	94.2	62.0
Gemini 3.8 Flash	95.3	69.7	57.1	80.0	89.7	94.0	64.5
GLM 5.3 Flash	92.8	57.0	71.1	78.6	82.9	91.6	57.9
Muse Spark 1.3	95.6	65.1	69.7	77.0	81.4	93.0	59.9
Doubao Seed 2.1 Pro	94.6	53.1	61.9	75.9	84.0	91.8	53.6

(a) Script tasks

	A1 Entities [79.5]	A2 Layout [23.4]	A3 Spaces [67.2]	P1 Picture [28.5]	B1 Routes [23.9]	B2 Events [54.0]	C1 Camera [52.1]	C2 Cuts [80.2]	D1 Physics [48.1]
GPT-6 Astra	81.4	**36.1**	73.9	**43.4**	**31.0**	**61.8**	61.8	**85.9**	54.0
GPT-6.1 Sol	80.9	33.9	75.8	41.6	29.6	60.9	60.8	85.0	53.4
Opus 5.5	**81.7**	27.3	75.5	40.1	25.3	59.2	58.3	83.2	**62.5**
Gemini 4 Argon	81.6	33.7	**77.0**	40.5	28.4	58.8	**62.9**	85.5	56.2
GPT-5.6 Sol	77.9	20.7	66.4	25.2	22.1	56.8	50.8	82.4	49.8
Kimi K3	79.5	21.3	61.1	21.7	23.8	54.7	47.5	74.7	43.9
DeepSeek V4.1 Flash	79.7	20.6	68.0	25.1	21.9	48.9	48.9	79.7	42.9
Gemini 3.8 Flash	77.9	18.2	64.7	22.5	21.4	49.0	48.3	83.5	47.2
GLM 5.3 Flash	78.1	16.9	61.2	19.0	18.6	53.2	45.7	70.9	40.6
Muse Spark 1.3	77.0	13.4	55.8	17.2	20.5	43.1	42.3	75.9	40.6
Doubao Seed 2.1 Pro	78.9	15.4	59.4	17.5	20.4	47.5	46.0	75.0	38.2

(b) Visual-input tasks

Figure 13. **Rule score per dimension** (failed cells count 0; column means in brackets; best per column in bold). (a) Script tasks. (b) Visual-input tasks (photos, continuations, videos). Rows sorted by rule score over all tasks.

and a blind recap retells only 56% of the story. Weaker agents thus err more and also make their correct parts harder to read. Chaining the two scores per event (Fig. 14), only 36% of annotated events are both accepted by the rules and confirmed on screen: about half for the four strongest agents (44–54%) and a fifth for the four weakest, the widest gap of any error type. A further 14% pass the rules but are rejected by the judge, so the rules alone would overstate event completion. Conversely, the rules miss some posture events such as falls, which lowers the confirmed share.

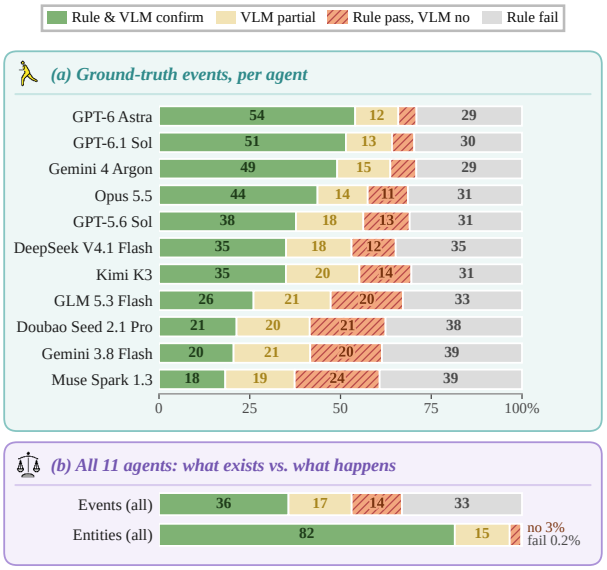


Figure 14. **Event completion.** Each ground-truth fact is triangulated by both scorers: the rules pass it and the VLM confirms it on screen, confirms it partly, or rejects it; or the rules already fail it. Facts without a usable VLM verdict are excluded. (a) Events per agent: confirmed events fall from 54% (GPT-6 Astra) to 18% (Muse Spark 1.3), and weaker agents more often stage an event the rules accept but the screen does not show. (b) Pooled over agents, entities almost never fail, while events do.

C.3 Do proxy errors reach the video?

For two script tasks we give Seedance 2.5 the script together with the proxy of a good and of a bad agent (Fig. 2). The generator paints appearance on top of the proxy but keeps its camera, framing and blocking: it cranes up only when the proxy does, and it turns the bad proxy's cylinder figure into a man without legs and its camera inside a wall into a blank wall. An agent's camera errors, the most systematic ones in Finding 2, therefore become the video's. Correlating benchmark scores with generated-video quality across all agents is left to future work.

C.4 Agent statistics on script tasks

Table 5. Per-task means on the 94 script tasks: turns, tool calls and wall-clock minutes, and the number of failed runs (no valid scene after re-execution, counted as in Tab. 2). Agents in the order of Tab. 2.

Agent	Turns	Tools	Min.	Failed
GPT-6 Astra	57	56	73	0
GPT-6.1 Sol	48	47	82	0
Gemini 4 Argon	28	62	67	2
Opus 5.5	38	40	57	0
GPT-5.6 Sol	85	84	41	0
DeepSeek V4.1 Flash	98	98	72	0
Kimi K3	78	96	118	0
Gemini 3.8 Flash	104	101	64	0
GLM 5.3 Flash	70	88	74	1
Doubao Seed 2.1 Pro	100	114	86	0
Muse Spark 1.3	70	91	88	0

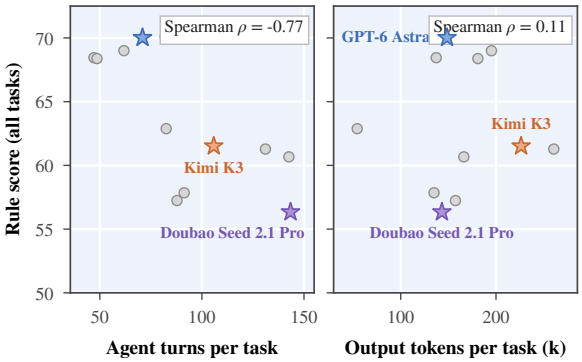


Figure 15. **Effort vs. score.** Mean turns and output tokens per task against the rule score (all tasks), one marker per agent (stars: best, median and weakest); Spearman ρ over the 11 agents.